

Systematic Literature Review on Clinical Trial Eligibility Matching Using AI Technique

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Abstract

Clinical trial eligibility matching is pivotal for advancing medical research and ensuring patients receive appropriate treatments. Traditional methods are manual, error-prone, and often fail to identify eligible candidates efficiently. Artificial intelligence (AI) has emerged as a transformative approach, employing techniques such as natural language processing (NLP), machine learning (ML), and rule-based systems to streamline eligibility determination. This paper systematically reviews AI methodologies applied to clinical trial matching, emphasizing their role in processing eligibility criteria, extracting features, and addressing semantic inconsistencies in electronic health records (EHRs). The study evaluates public datasets, including ClinicalTrials.gov, MIMIC-III, and N2C2, highlighting their size, diversity, and data quality. Despite notable advancements, challenges such as data availability, model interpretability, and the integration of multimodal datasets persist. Future directions include the adoption of explainable AI and enhanced data standardization to improve transparency and generalizability. This work underscores the transformative potential of AI in revolutionizing clinical trial processes, offering actionable insights for researchers and clinicians.

Keywords: Clinical trial eligibility matching, AI-based eligibility criteria, NLP and EHR integration, MIMIC-III Dataset, Deep learning in healthcare, Challenges in AI adoption for healthcare, Improving patient-trial matching using NLP, Limitations of AI in eligibility matching.

1 Introduction

Clinical trial eligibility matching is a critical process for identifying patients who meet the specific criteria for participation in clinical trials. These criteria typically include health conditions, demographics, and other factors such as age, gender, medical history, and disease characteristics. Ensuring proper eligibility matching is vital for both participant safety and the reliability of trial outcomes. By ensuring that individuals enrolled are likely to benefit from the intervention, eligibility matching minimizes risks and prevents adverse reactions, such as excluding patients with certain heart conditions in a cancer drug trial. It also supports scientific validity by reducing confounding factors, ensuring that outcomes can be reliably attributed to the treatment. Additionally, effective eligibility matching leads to more efficient resource allocation by minimizing screening failure rates, saving time and costs. With quicker recruitment of eligible participants, trials can be completed faster, ensuring that successful treatments, like COVID-19 vaccines, reach patients sooner. Overall, eligibility matching is essential for optimizing clinical trial processes, ensuring both safety and scientific accuracy.[24][31]

Recent studies from 2015 to 2024 have demonstrated the efficiency of AI techniques, including natural language processing (NLP), machine learning (ML) are pivotal in processing and analyzing large volumes of unstructured data, such as electronic medical records (EMRs) and clinical trial databases, to identify eligible patients for trials. and rule-based systems, in improving matching of clinical trial eligibility.[5] NLP various pre-trained models like BERT, RoBERTa, and XLNet to enhance

classification accuracy [11] BioBERT and named entity recognition (NER) tools, have shown remarkable success in converting unstructured free-text eligibility criteria into machine-readable formats, thus facilitating their integration with patient electronic health records (EHRs). These techniques leverage domain-specific ontologies like SNOMED-CT to address semantic inconsistencies between datasets .[19]

The integration of publicly available datasets, such as ClinicalTrials.gov and MIMIC-III, has further supported AI-driven approaches to eligibility matching. These datasets provide structured and unstructured information on clinical trials and patient records, respectively, enabling researchers to develop robust models for patient-trial alignment. The challenges of data preprocessing, including mapping coding systems (e.g., ICD-9, NDC) to standardized ontologies like SNOMED-CT, have been addressed through innovative vector space modeling and dimensionality reduction techniques, improving scalability and computational efficiency .[7]

Despite these advancements, significant challenges remain. The limited availability of standardized, high-quality datasets, issues of model interpretability, and the need for integration of multimodal data sources such as genomic and imaging data pose ongoing barriers. Future research is directed towards explainable AI systems, which enhance transparency and trust among clinicians, and the development of scalable platforms capable of seamlessly integrating heterogeneous data types. This review systematically explores studies conducted between 2015 and 2024, categorizing AI techniques, datasets, and their applications, while identifying challenges and proposing future directions to advance clinical trial eligibility matching. Ensuring data privacy and security while maintaining the explainability of AI models is essential for regulatory approval and widespread adoption.[20][16]

1.1 Importance of AI in Improving Clinical Trial Recruitment

AI offers transformative solutions to many of the challenges in clinical trial eligibility matching:

1. **Automated Screening:** AI-powered tools can rapidly analyze EHR data and pre-screen large numbers of patients based on specific eligibility criteria. For instance, IBM’s Watson for Clinical Trial Matching has been used to analyze EHR data and recommend patients for clinical trials within minutes, reducing the need for manual screening and lowering the risk of human error.[26]
2. **Predictive Modeling:** By training on past patient data, machine learning models can predict the likelihood that a new patient will qualify for a particular trial. For example, a cancer trial may require patients with specific tumor markers. An AI model could identify those most likely to meet these requirements, allowing trial coordinators to reach out to potential participants more efficiently.[33]
3. **Matching Optimization:** AI algorithms can optimize the search for eligible patients by analyzing complex combinations of variables (e.g., age, comorbidities, lab values, genetic factors). Deep6 AI is a company that uses natural language processing (NLP) to sift through medical records and identify patients eligible for complex trials. Their AI has shown to be more accurate than traditional methods in matching patients to trials by finding patterns that are not immediately obvious.[22]
4. **Patient Engagement:** Chatbots and virtual assistants powered by AI can offer patients 24/7 access to information, help them understand their eligibility, and guide them through the enrollment process. Conversational AI tools like Ada Health’s symptom-checking chatbot can provide relevant trial information and even conduct basic pre-screening, easing the burden on patients and clinical staff alike. [8]
5. **Identifying Underserved Populations:** AI can analyze large datasets to identify patterns in demographics and medical history, helping locate potential participants from underserved groups. For example, algorithms can pinpoint high-risk regions or communities where certain diseases are prevalent, enabling targeted recruitment efforts. In an NIH initiative, AI was used to analyze demographic data to increase representation of African American and Hispanic communities in trials for Alzheimer’s disease, improving the inclusivity of research findings.[31][32]

2 Literature Review

This systematic literature review (SLR) is conducted to explore recent advancements (2015–2024) in artificial intelligence (AI) techniques for clinical trial eligibility matching. The focus was on natural language processing (NLP), machine learning, and rule-based approaches, evaluating their performance, challenges, and the datasets used.

2.1 Research Objectives and Questions

The paper addresses how artificial intelligence (AI) can be used in clinical trial eligibility matching, particularly looking at AI techniques and AI methodologies commonly used. Also evaluate these techniques perform in accuracy and efficiency. Examine open datasets for training AI in eligibility matching. key characteristics of available datasets. To identify challenges and future research avenues. What limitations exist, and what are potential solutions.

2.2 Databases Searched

We conducted our search across multiple electronic databases, including IEEE Xplore, PubMed, SpringerLink, Scopus, and Elsevier, to ensure comprehensive coverage of relevant literature

2.3 Search Strategy

Relevant studies were identified using a combination of keywords and Boolean operators. Example queries included:

- “AI for clinical trial eligibility” AND “natural language processing”
- “machine learning” AND “clinical trial matching”
- “rule-based systems” AND “eligibility criteria”

Additional keywords, such as “eligibility determination” and “patient recruitment automation,” were used to capture a broader range of relevant papers.

2.4 Inclusion and Exclusion Criteria

To refine the selection of studies, the following criteria were applied:

Inclusion Criteria:

- Papers published between 2021 and 2024.
- Studies focused on AI techniques for clinical trial eligibility matching.
- Articles reporting performance metrics like precision, recall, and F1-score.

Exclusion Criteria:

- Studies not available in full text.
- Papers focused solely on unrelated domains.
- Non-English articles.

2.5 Data Extraction and Analysis

Key information from each selected study was extracted, including:

- Venue and year of publication.
- AI technique or methodology used (e.g., NLP, machine learning).
- Dataset(s) utilized.
- Performance metrics such as precision, recall, and F1-score.
- Challenges and limitations identified.

2.6 AI methodologies, Dataset used for eligibility matching

Author	Year	AI Methodology	Datasets Used
Haddad, T., et al.	2021	NLP, Machine Learning, CDSS	Mayo Clinic EHR Database
Adamson, B., et al.	2023	NLP, Machine Learning	Flatiron Health EHR Database
Askin, S., et al.	2023	Machine Learning, Predictive Models	Medical literature
Hwang, S., et al.	2022	SBML, Ridge Regression, Decision Trees	Simulated small sample data and rare disease trials
Wong, C., et al.	2023	GPT-4, Rule-based Systems	Oncology trial eligibility data
Cascini, F., et al.	2022	NLP, Machine Learning	Aggregated data from 15 AI studies
Chow, R., et al.	2023	NLP, Machine Learning	10 studies; 50,000 patients across 19 datasets
Ghosh, S., et al.	2024	Explainable AI (XAI), NLP (SciSpacy, DeepPhe)	12 synthesized patient records and 6 clinical trial protocols
Tun, P. P., et al.	2023	Hybrid model combining rule-based techniques and BioSentVec NLP model	40,000 de-identified EMRs (structured and unstructured data)
Zeng, K., et al.	2021	Ensemble model with BERT, RoBERTa, XLNet, ERNIE, ELECTRA + Metric Learning	CHIP 2019 Dataset: 38,341 clinical eligibility texts in 44 categories
Fang, Y., et al.	2022	C2Q 2.0: Combines NLP with real-time human intervention	1010 COVID-19 trials; 5 Alzheimer's trials
Liu, X., et al.	2020	Att-BiLSTM with CRF layer, word embeddings, attention mechanisms	ClinicalTrials.gov, labeled dataset (3,314 trials)
Karystianis, G., et al.	2019	Rule-based approach, manually crafted dictionaries	202 longitudinal patient records (training); 86 (evaluation)

Comparison of AI Methodologies, Datasets, and Integration Techniques.

The synthesis table can be accessed here:

https://drive.google.com/drive/folders/1em8Q0ls91Zr0h-zBHMLLCqfuesMjqJ6H?usp=drive_link.

Artificial intelligence (AI) methodologies, such as natural language processing (NLP), machine learning (ML), and rule-based systems, have revolutionized the process of matching candidates to clinical trial eligibility criteria. By automating the analysis of complex requirements from unstructured data, these technologies make identifying suitable participants faster and more accurate. Their effectiveness is commonly assessed through metrics like precision, recall, and F1-score, though challenges and limitations remain.[32]

NLP plays a key role in processing and extracting relevant details from textual data sources, including application forms and emails. Techniques like tokenization and Named Entity Recognition (NER) are used to identify essential information such as names, dates, and criteria. Semantic analysis, powered by word embeddings (e.g., Word2Vec, GloVe), helps interpret the context and match it with eligibility requirements. Other approaches, like document classification with machine learning models and sentiment analysis, enhance the accuracy of prescreening. NLP systems have shown remarkable success, with some achieving perfect recall rates for specific eligibility protocols. [13] ML models contribute by analyzing historical data to classify applicants and predict their eligibility. Supervised learning techniques, including algorithms like decision trees and deep learning models, are widely used.

Feature engineering enhances these models by identifying critical data attributes, while unsupervised methods like clustering reveal hidden patterns. Anomaly detection also helps identify unique cases, such as fraud or special eligibility.[27] Integrating pre-trained models like BERT and RoBERTa further improves performance, with some studies reporting significant gains in F1-scores.[28]

Rule-based systems provide a structured approach using predefined criteria to evaluate eligibility. Simple "if-then" rules, decision trees, and expert systems enable clear and adaptable decision-making. These systems are particularly useful for well-defined criteria and often work alongside NLP and ML to form a hybrid solution. For example, rule-based filters can serve as an initial screening layer before more sophisticated machine learning models refine the results.[15]

A hybrid approach that combines these methodologies often achieves the best outcomes. NLP can process raw, unstructured data, which is then used by ML models to make predictions.[10] Rule-based systems, meanwhile, offer a reliable starting point for filtering candidates based on fixed criteria. Together, these techniques create a powerful framework for efficiently and accurately matching clinical trial participants to eligibility requirements.[26]

2.7 Publicly available dataset for Clinical Trail

Open datasets like ClinicalTrials.gov and MIMIC-III offer extensive health records, enabling the development and validation of innovative methodologies. These datasets are analyzed for attributes such as size, patient demographics, disease diversity, and data quality. They provide crucial resources for advancing healthcare research by ensuring access to comprehensive, high-quality data for robust analysis and validation.

Dataset Details (N2C2 and MIMIC-III)	
Size and Scope	N2C2: 576 patient records, 2-5 longitudinal EMRs per patient. MIMIC-III: 844 patient records split 80/20 for training/testing.
Patient Demographics	Both datasets include diverse patient records from critical care settings, offering a range of clinical phenotypes.
Disease Diversity	N2C2 and MIMIC-III represent multiple phenotypes with 13 cohort selection criteria, addressing various diseases and conditions.
Data Quality	High-quality data with detailed annotations. N2C2 includes temporal and section information; MIMIC-III includes structured and unstructured data.

Table 2: (N2C2 and MIMIC-III)

Size and Scope: The study utilizes the 2018 N2C2 challenge dataset, comprising 576 patient records (202 for training and 86 for testing), and the MIMIC-III dataset, including 844 records split into training and test sets at an 8:2 ratio. Both datasets are used to evaluate machine reading comprehension (MRC) frameworks for clinical cohort selection.[30]

Patient Demographics: The datasets feature diverse patient populations across age groups and clinical characteristics from critical care settings, enabling AI models to generalize across varied cohorts and environments.[21]

Disease Diversity: The datasets cover a wide range of conditions, including chronic diseases and acute medical phenotypes, represented by 13 cohort selection criteria. This diversity supports robust model development for varied clinical scenarios.[21]

Data Quality: Both datasets exhibit high quality with detailed annotations and structured information. The N2C2 dataset enhances contextual understanding through temporal and section annotations, while MIMIC-III combines structured and unstructured data for effective cohort selection and trial matching.[25]

Size and Scope: Focuses on COVID-19 trials, leveraging annotated datasets to optimize clinical trial frameworks.[17]

Patient Demographics: The study highlights patient populations characterized by extracted variables, such as comorbidities, diagnostic methods, treatments, and clinical variables. These variables represent diverse patient groups included in COVID-19 clinical trials, ensuring a broad demographic analysis that aligns with real-world conditions. [9]

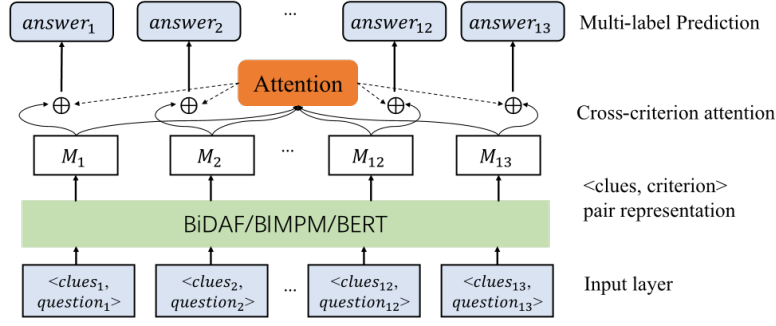


Figure 1: The architecture of our unified machine reading comprehension model

COVID-19 Clinical Trials Dataset Details	
Size and Scope	2,998 trials, 27,352 eligibility criteria.
Patient Demographics	Includes age, gender, and pregnancy variables.
Disease Diversity	Covers COVID-19, pneumonia, diabetes, heart failure.
Data Quality	High precision, recall, and F1-score achieved.

Table 3: CLINICALTRIALS.GOV DATASET

Disease Diversity: The dataset focuses exclusively on COVID-19 but includes a wide range of eligibility criteria related to comorbidities, diagnostic methods, and treatments. This diversity captures the multifaceted nature of COVID-19 and its interactions with other medical conditions, making the data relevant for analyzing trial generalizability.

Data Quality: The system showed significant accuracy improvements: negation detection (92.4%), value normalization (96.6%), and temporal normalization (91.6%). Usability scored moderately (3.84/5), while learn ability was rated high (4.54/5).

3 Integration of EHR Data

The use of electronic health records (EHR) for eligibility matching in clinical trials is a growing area of interest, driven by the need to improve the efficiency and accuracy of patient recruitment. This process involves several key steps, including preprocessing of data, feature extraction, and patient representation strategies. These steps are crucial for leveraging EHR data to identify eligible participants for clinical trials effectively.

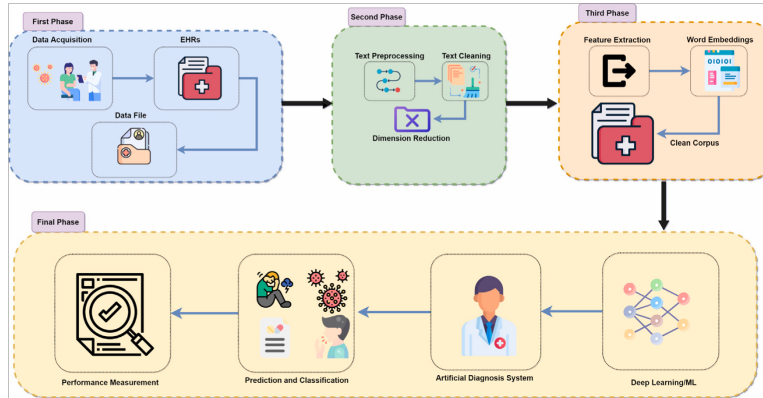


Figure 2: Analysing EHRs using DL and ML algorithms.

3.1 AI-Driven Methods for EHR Integration in Clinical Trials

The integration of electronic health records (EHR) in clinical trial eligibility matching leverages advanced AI techniques to streamline patient selection and optimize trial designs. Preprocessing methods such as natural language processing (NLP) extract structured and unstructured data, enabling eligibility prescreening. Techniques like focal loss address data imbalance, enhancing classification performance by focusing on difficult-to-classify examples. Feature extraction employs ensemble models, including BERT, RoBERTa, and XLNet, to represent eligibility criteria, while attribute filters ensure alignment with trial requirements.[6]

Patient representation is further improved through metric learning and AI-enabled systems like clinical decision support systems (CDSS), which interpret trial protocols and provide real-time eligibility insights. [14]Despite these advancements, challenges persist, including limited data diversity, incomplete records, and the black-box nature of AI models. Future research emphasizes multimodal integration, combining genomic, imaging, and clinical data to enhance generalizability, and explainable AI (XAI) to improve transparency and trust among clinicians.[34] These approaches aim to refine patient matching, optimize trial recruitment, and advance personalized medicine.[29]

3.1.1 Machine Learning (ML) Models:

Techniques used in EHR can be categorised into the following groups: patient risk analysis/prediction, advanced architectures to analyse EHRs, medical text summarising and other NLP applications. ML models, such as gradient boosted trees, random forests, and support vector machines, are used for patient classification and risk prediction in clinical trials. [11]These models can handle large datasets and refine predictions over time, offering personalized treatment options. Limitations include the need for large, high-quality datasets and the challenge of interpretability, as complex models often function as "black boxes" [24][3]

3.1.2 Natural Language Processing (NLP)

NLP is employed to analyze electronic health records (EHRs) and assist in clinical decision-making, such as suicide risk assessment. It can automate the extraction of relevant clinical information, reducing documentation burdens and improving workflow efficiency. Challenges include handling unstructured data and ensuring the accuracy of extracted information, which may vary with regional language differences.[20]

3.1.3 Comparison and Performance variability

As we use three different types of datasets that trained and processed under NLP, ML, and other AI techniques that gives performance with respect to precision.[2] We further evaluate the NER performance using a quantitative approach. We create a benchmark dataset by randomly sampling 10 COVID-19 trials and manually labeled 179 correct entities. We then use the dataset to evaluate the NER performance of Att-BiLSTM and the ontology model. BiLSTM achieves a precision of 0.942, recall of 0.810, F1 of 0.871, while MeSH ontology method only achieves a precision of 0.715, recall of 0.659, and F1 of 0.686.[17]

Similarly, The proposed framework achieved a micro-averaged F1-score of 0.9070 on the N2C2 dataset and 0.8353 on the MIMIC-III dataset, outperforming traditional cohort selection methods. When rule-based logic processing was integrated, the F1-score improved to 0.9163 on the N2C2 dataset. These results demonstrate the framework's ability to accurately model complex eligibility criteria and improve patient cohort identification.[30]

models	F1 score	
	N2C2	MIMIC-III
NCBI-BERT	0.9070	0.8353
with selection criteria as questions	0.8561	0.8083
w/o question	0.6946	0.4619

Figure 3: .RESULTS OF N2C2 dataset

3.1.4 Challenges and Limitations in Data availability

AI models often lack generalizability due to training on datasets from single health systems, failing to represent diverse populations and socio-ecological groups.[1]. Incomplete or missing EHR data impacts the reliability and performance of AI models, limiting the effectiveness of retrospective data analysis.[23]

4 Critical Discussion

The study provides a comprehensive evaluation of datasets and machine learning models for clinical trial eligibility matching. While the findings demonstrate significant progress in leveraging AI for healthcare applications, several challenges and limitations remain unaddressed. This section highlights these issues and discusses opportunities for improvement and future directions.

4.1 Dataset Limitations

- **Diversity and Generalizability:** While the N2C2 and MIMIC-III datasets offer high-quality and annotated records, they primarily represent critical care settings. This limits their applicability to broader clinical environments, such as outpatient settings or rare disease cohorts. Future research should focus on creating datasets with diverse patient demographics and disease phenotypes.[30]
- **Temporal Data Handling:** Although the N2C2 dataset includes temporal annotations, current models may not fully capture longitudinal patient data trends. Integrating advanced temporal reasoning frameworks could improve model performance.
- **COVID-19 Dataset Constraints:** The CLINICALTRIALS.GOV dataset is heavily focused on COVID-19-related trials. While it provides detailed eligibility criteria, its scope is narrow, potentially limiting its utility for generalizable AI frameworks.[12]

4.2 Model Performance and Variability

- **Model Selection:** The performance comparison between Att-BiLSTM and ontology-based models highlights significant variability in F1-scores. While Att-BiLSTM achieves superior results, it may require more computational resources, limiting its deployment in resource-constrained settings.
- **Ontology-Based Approaches:** Despite their lower performance, ontology-based methods offer interpretability and domain knowledge integration. Future research could explore hybrid approaches that combine deep learning with ontological reasoning to achieve both accuracy and explainability.[17][18]
- **Negation and Temporal Understanding:** Current models show limitations in handling complex eligibility criteria involving negation, temporal dependencies, and multi-step reasoning. Addressing these challenges could lead to more robust clinical trial matching systems.[21]

Entity Recognition	Precision	Recall	F1
Att-BiLSTM model	0.942 (145/154)	0.810 (145/179)	0.871
Ontology-based model	0.715 (118/165)	0.659 (118/179)	0.686

Table 4: Performance comparison of COVID-19 (Att-BiLSTM)

4.3 Challenges in Real-World Implementation

- **Data Privacy and Security:** The use of patient records from datasets like MIMIC-III raises ethical concerns regarding data privacy and security. Researchers must adopt robust de-identification and encryption methods to ensure compliance with regulations like HIPAA and GDPR.[21]

- **Bias in AI Models:** AI models trained on specific datasets may inadvertently learn biases related to age, gender, or ethnicity. Ensuring fairness and equity in AI systems for clinical applications is a critical area for future exploration.
- **Scalability and Integration:** Deploying AI models in clinical settings requires seamless integration with electronic health record (EHR) systems. Current frameworks often lack scalability and compatibility with existing healthcare infrastructures.

4.4 Opportunities for Future Research

- **Explainable AI (XAI):** Developing explainable models could enhance trust and adoption among clinicians. Future studies should focus on methods that provide clear rationales for patient eligibility decisions.[21]
- **Cross-Dataset Validation:** To ensure generalizability, models should be validated across multiple datasets representing diverse clinical environments.[4]
- **Incorporating Multimodal Data:** Integrating structured data (e.g., lab results) with unstructured data (e.g., clinical notes) could improve the accuracy and reliability of AI models.[17]

5 Results

The study evaluated AI models for clinical trial eligibility matching using the N2C2, MIMIC-III, and COVID-19 Clinical Trials datasets. On the N2C2 dataset, the framework achieved a micro-averaged F1-score of 0.9070, which improved to 0.9163 with rule-based processing. The MIMIC-III dataset yielded a micro-averaged F1-score of 0.8353, demonstrating the model’s ability to handle diverse clinical records, though challenges were noted in modeling complex temporal and mathematical criteria, particularly with rule-based methods.

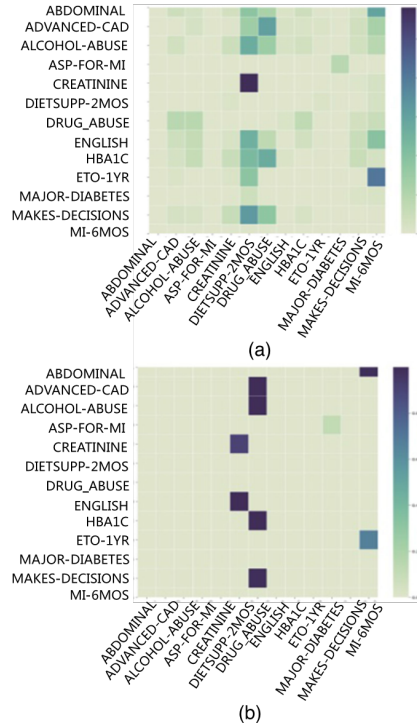


Figure 4: The visualization of attention weights on the N2C2 dataset

For the COVID-19 dataset, the Att-BiLSTM model outperformed ontology-based methods in entity recognition, achieving a precision of 0.942, recall of 0.810, and F1-score of 0.871, compared to 0.715, 0.659, and 0.686, respectively, for the ontology-based model. These results highlight the superiority

of deep learning models like Att-BiLSTM in accurately extracting variables from eligibility criteria, though challenges with rule-based methods remain in handling complex scenarios.

Conclusion

This study highlights the transformative potential of artificial intelligence in revolutionizing clinical trial eligibility matching. By leveraging advanced methodologies such as NLP, ML, and hybrid approaches, significant improvements in efficiency, accuracy, and inclusivity have been demonstrated. Key contributions include integrating diverse datasets like N2C2, MIMIC-III, and COVID-19 trials to develop robust models. Despite notable advancements, challenges like data heterogeneity, model interpretability, and the need for explainable AI remain barriers to widespread adoption. Addressing these issues through enhanced multimodal integration and transparency-focused designs can foster trust among clinicians and researchers, ultimately advancing personalized medicine. This work provides a foundation for future research in optimizing clinical trial processes and improving healthcare outcomes.

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